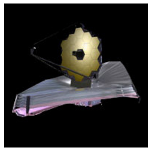

6927 - Groundbreaking Constraints on the Galactic Dust Extinction Law in the High-Opacity Regime

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NIRSpec MSA Spectra				
	22	Brick Pointing 1 of 4 (Obs 22 -- medium version)	NIRSpec MultiObject Spectroscopy	(22) MPT_sourcelist.v22.all
	19	Brick Pointing 2 of 4, screened catalog	NIRSpec MultiObject Spectroscopy	(23) MPT_sourcelist.v22.pointing2

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	28	Brick Pointing 4 of 4, screened catalog, 3x3 replan	NIRSpec MultiObject Spectroscopy	(25) MPT_sourcelist.v22.all-22-19-23
	23	Brick Pointing 3 of 4, screened catalog	NIRSpec MultiObject Spectroscopy	(24) MPT_sourcelist.v22.all.-22-19

ABSTRACT

Given its fundamental importance for so many areas of astrophysics, the Galactic dust extinction curve has attracted a great deal of attention, and is already well-characterized in environments where the extinction is relatively low (of order Av<5 mag). Unfortunately, the extinction curve remains poorly constrained in the high-opacity regime critical for understanding grain growth, the ice content of dense molecular clouds, molecular cloud structure, the 3D structure of the Galaxy, and many other phenomena. To help remedy this situation, we seek NIRSpec MSA spectra prioritizing highly obscured stars in a well-known infrared dark cloud, the Brick. Cycle 1 NIRCam imaging of the Brick has revealed thousands of highly obscured stars in and around this extremely dense object at Av's up to 50 mag, ideal for characterizing the high-opacity dust extinction curve with groundbreaking precision. As our fits to hundreds of simulated attenuated stellar spectra demonstrate, our proposed MSA spectra of over 200 stars will successfully constrain the 1 to 5 micron spectral dust extinction curve over a wide range of dust extinctions. With an abundance of targets, our proposed observations will provide transformative constraints on the spectroscopic extinction curve as a function of Av, an outcome with wide applications, and one not possible from any other facility. NIRSpec's high spectral sensitivity and resolving power are needed for this work: it cannot be done with broadband photometry.

OBSERVING DESCRIPTION

We propose NIRSpec MSA spectra covering the full NIRSpec bandpass from 1-5 um, targeting obscured stars in the Brick, an extremely dense infrared dark cloud located near the Galactic center. The program will yield groundbreaking measurements of the Galactic dust extinction law at extreme opacities in a unique environment only accessible to JWST.

The observations are designed to cover the entire spatial extent of the Brick, so as to sample all existing environments (all available dust opacities). For that, four pointings of the MSA footprint must be tiled to cover the chip gaps. At each pointing, we seek to obtain dithered spectra in three grating settings: G140M/F100LP, G235M/F270LP, and G395M/F290LP, to obtain the desired spectral coverage. This is necessary to break the degeneracies between intrinsic stellar colors and reddening imposed by the dust within the Brick. We use medium spectral resolution for efficiency. The exposure time is set by the need to have adequate SNR per NIRSpec spectral channel for our faintest targets.

The Brick will be available during two ~50-day windows in 2025 Aug/Sep and 2026 Mar-May, at position angles of roughly 230 and 50 degrees, respectively. Only the 2025 Aug/Sep window meets the micrometeoroid avoidance requirement, so we adopt 230 degrees. This serendipitously aligns the NIRSpec MSA diagonal axis with the Brick's long axis.

The astrometry of our targets is better than 0.03", based on multiband Cycle 1 NIRCam imaging (PID 2221).

We use the nod-in-slitlet strategy for background subtraction and to improve the spectral SNR. This will yield clean spectra because our targets are selected not to have near neighbors.

The four MSA pointings yield 53, 53, 53, and 54 spectra of unique targets, or 213 spectra in total.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(20)	MPT_sourcelist.v22	RA: 17 46 7.8250 (266.5326042d) Dec: -28 43 43.45 (-28.72874d) Equinox: J2000		
Comments: *Description=[]*				
(21)	MPT_sourcelist.v22.filler	RA: 17 46 8.4578 (266.5352408d) Dec: -28 42 8.81 (-28.70245d) Equinox: J2000		
Comments: *Description=[]*				
(22)	MPT_sourcelist.v22.all	RA: 17 46 8.3278 (266.5346992d) Dec: -28 42 24.39 (-28.70677d) Equinox: J2000		
Comments: *Description=[]*				
(23)	MPT_sourcelist.v22.pointing2	RA: 17 46 8.3304 (266.5347100d) Dec: -28 42 24.04 (-28.70668d) Equinox: J2000		
Comments: *Description=[]*				
(24)	MPT_sourcelist.v22.all.-22-19	RA: 17 46 8.3177 (266.5346571d) Dec: -28 42 23.77 (-28.70660d) Equinox: J2000		
Comments: *Description=[]*				
(25)	MPT_sourcelist.v22.all-22-19-23	RA: 17 46 8.3177 (266.5346571d) Dec: -28 42 24.04 (-28.70668d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 6927, Observation 22: Brick Pointing 1 of 4 (Obs 22 -- medium version) Mon Jun 08 20:00:55 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(22)	MPT_sourcelist.v22.all	RA: 17 46 8.3278 (266.5346992d) Dec: -28 42 24.39 (-28.70677d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	MPT_sourcelist.v22.all (7566 sources)	MPT_sourcelist.v22.all (7566 sources)	jwst-nirspec-mr	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	266.537983625 Degrees -28.728375555555544 Degrees	229.99846710260877			3	6	1838.2
2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	266.537983625 Degrees -28.728375555555544 Degrees	229.99846710260877			3	6	1838.2
3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	266.537983625 Degrees -28.728375555555544 Degrees	229.99846710260877			3	6	1838.2

Special Requirements	
	No Parallel Attachments MSA Planned Aperture PA 230.0000 to 230.0000 Degrees (V3 91.4254303 to 91.4254303)

Observation

Proposal 6927, Observation 19: Brick Pointing 2 of 4, screened catalog

Mon Jun 08 20:00:55 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 19:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(23)	MPT_sourcelist.v22.pointing2	RA: 17 46 8.3304 (266.5347100d) Dec: -28 42 24.04 (-28.70668d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	MPT_sourcelist.v22.pointing2 (7504 sources)	MPT_sourcelist.v22.pointing2 (7504 sources)	jwst-nirspec-mr	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	266.5414014583333 Degrees -28.717296944444456 Degrees	229.99680767992066			3	6	1838.2
2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	266.5414014583333 Degrees -28.717296944444456 Degrees	229.99680767992066			3	6	1838.2
3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	266.5414014583333 Degrees -28.717296944444456 Degrees	229.99680767992066			3	6	1838.2

Special Requirements	
	No Parallel Attachments MSA Planned Aperture PA 230.0000 to 230.0000 Degrees (V3 91.4254303 to 91.4254303)

Observation

Proposal 6927, Observation 28: Brick Pointing 4 of 4, screened catalog, 3x3 replan Mon Jun 08 20:00:55 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 28:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(25)	MPT_sourcelist.v22.all-22-19-23	RA: 17 46 8.3177 (266.5346571d) Dec: -28 42 24.04 (-28.70668d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	MPT_sourcelist.v22.all-22-19-23 (7394 sources)	MPT_sourcelist.v22.all-22-19-23 (7394 sources)	jwst-nirspec-mr	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	266.53597129166667 Degrees -28.708379166666646 Degrees	229.99937218273422			3	6	1838.2
2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	266.53597129166667 Degrees -28.708379166666646 Degrees	229.99937218273422			3	6	1838.2
3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	266.53597129166667 Degrees -28.708379166666646 Degrees	229.99937218273422			3	6	1838.2

Special Requirements	
	No Parallel Attachments MSA Planned Aperture PA 230.0000 to 230.0000 Degrees (V3 91.4254303 to 91.4254303)

Observation

Proposal 6927, Observation 23: Brick Pointing 3 of 4, screened catalog

Mon Jun 08 20:00:55 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(Visit 23:1) Warning (Form): Fewer than 5 usable reference stars were found for this visit.

(Visit 23:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(24)	MPT_sourcelist.v22.all.-22-19	RA: 17 46 8.3177 (266.5346571d) Dec: -28 42 23.77 (-28.70660d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	MPT_sourcelist.v22.all.-22-19 (7450 sources)	MPT_sourcelist.v22.all.-22-19 (7450 sources)	jwst-nirspec-mr	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	266.5376949166666 Degrees -28.702851944444433 Degrees	229.99853289171517			3	6	1838.2
2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	266.5376949166666 Degrees -28.702851944444433 Degrees	229.99853289171517			3	6	1838.2
3	3 (G395M/F290LP)	c1	3 Shutter Slitlet	266.5376949166666 Degrees -28.702851944444433 Degrees	229.99853289171517			3	6	1838.2

Special Requirements	
	No Parallel Attachments MSA Planned Aperture PA 230.0000 to 230.0000 Degrees (V3 91.4254303 to 91.4254303)